

Physiology of reproductive system

The functions of the female reproductive system

1. formation of female gametes, ova
2. reception of male gametes, spermatozoa
3. provision of suitable environments for fertilization of the ovum by spermatozoa and development of the resultant fetus
4. parturition (childbirth)
5. lactation

Puberty in the female

- Puberty is the age at which the internal reproductive organs reach maturity.
- The age of puberty varies between 10 and 14 years

The physical and psychological changes take place at this time is:

1. the uterus, the uterine tubes and the ovaries reach maturity
2. the menstrual cycle and ovulation begin
3. the breasts develop and enlarge
4. pubic and axillary hair begins to grow
5. there is an increase in the rate of growth in height and widening of the pelvis
6. there is an increase in the amount of fat deposited in the subcutaneous tissue, especially at the hips and breasts.

Functions of the ovaries

- Maturation of the follicle is stimulated by follicle stimulating hormone (FSH) from the anterior pituitary, and estrogen secreted by the follicle lining cells.
- **Ovulation** is triggered by a surge of luteinizing hormone (*LH*) from the anterior pituitary, which occurs a few hours before ovulation.
- After ovulation, the follicle lining cells develop into the corpus luteum (yellow body), under the influence of *LH* from the anterior pituitary.
- The corpus luteum produces the hormone *progesterone* and some *estrogens*.
- If the ovum is *fertilized* it embeds itself in the wall of the uterus where it grows and develops and produces the hormone **human chorionic gonadotrophin (hCG)**.
- *HCG* stimulates the corpus luteum to continue secreting *progesterone* and *estrogen* for the first 3 months of the pregnancy, after which time this function is continued by the *placenta*.
- If the ovum is *not fertilized* the corpus luteum degenerates and a new cycle begins with menstruation.

- At the site of the degenerate corpus luteum an inactive mass of fibrous tissue forms, called the *corpus albicans*.
- Sometimes more than one follicle matures at a time, releasing two or more ova in the same cycle.
- When this happens and the ova are fertilized the result is a multiple pregnancy.

FUNCTIONS OF OVARIAN HORMONES

- **Prepare for reproduction by:**
 1. Maturing the primordial germ cells
 2. Developing tissues to allow implantation of blastocyst
 3. Provides hormonal timing for ovulation
 4. Prepares environment needed for pregnancy
 5. Influences parturition and lactation
- **Under estrogen stimulation:**
 1. Vaginal epithelium proliferates
 2. Uterine endothelium proliferates
 3. Uterine glands hypertrophy
 4. Myometrium develops an intrinsic, rhythmic motion
 5. Breast ducts proliferate
 - Estradiol has anabolic effects on bone and cartilage
 - Estrogen causes vasodilation and heat dissipation

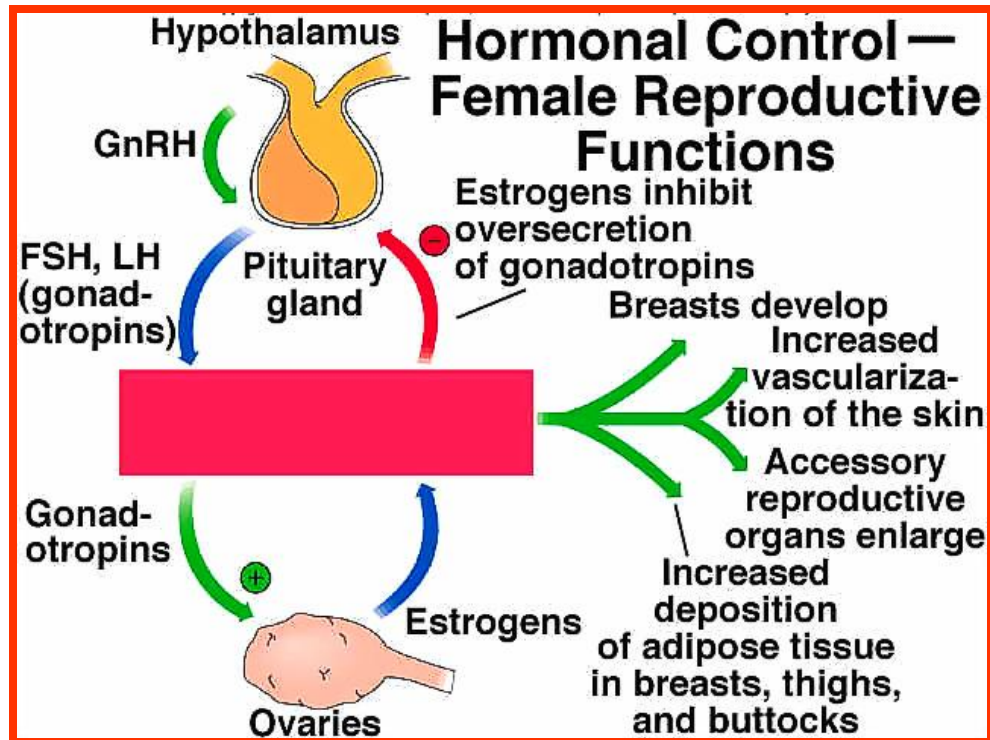
EFFECT OF PROGESTINS

1. Reduce proliferation of vaginal epithelium
2. Convert uterine epithelium from proliferative to secretory
3. Prepare for implantation of zygote
4. Enhance development of acinar portion of breast glands
5. Decrease peripheral blood flow resulting in increased body temperature

Hormonal control of ovaries.

These hormones include:

1. The hypothalamus produces GnRH. GnRH stimulates the anterior pituitary to produce FSH and LH.
2. FSH responsible for maturation of follicles to reach mature follicle (Graafian follicle) and stimulates the follicle to produce *estrogen*,
3. LH stimulates the corpus luteum to produce *progesterone*.
4. Estrogen and progesterone maintain the sexual organs (e.g., uterus) and the secondary sex characteristics, and exert feedback control over the hypothalamus and the anterior pituitary.



▪ **Female sexual cycles:**

- Events that recur every month ; not pregnant, Average 28 days in length (25-35)
- Include ovarian cycle and menstrual (uterine) cycle

1- **Ovarian cycle:** Each ovarian cycle is composed of 3 phases:

- a) Follicular phase
- b) Ovulation phase
- c) Luteal phase

Follicular phase: characterized by:

- Day 1 to day 14
- high levels of FSH secreted
- FSH promotes the maturation of a follicle in the ovary to *Graafian* follicle
- Follicular cells secrete estrogen

Ovulation phase

- Ovulation is the release of ovum (secondary oocyte) from rupture *Graafian* follicle into the peritoneal cavity.
- It occurs at the 14th day of the cycle.
- LH hormone is essential for ovulation.

Luteal phase (15-28 day)

- The corpus luteum is formed during this phase.
- Its duration is about **14 days**.

- The formed corpus luteum secretes *progesterone* and *estrogens* hormones.

If pregnancy occurs: corpus luteum continues to develop and secrete till the **4th month** of pregnancy and is controlled by *placental human chorionic gonadotrophins*.

If no pregnancy occurs: corpus luteum (yellow body) begins to degenerate and stops secretion 3-4 days before uterine bleeding.

2- Menstrual (uterine) cycle: The menstrual cycle is divided into three phases;
Menstrual, Proliferative and Secretory phases.

- The menstrual cycle is the structural and functional changes that occur in the Endometrium every month, from puberty to menopause.
- These changes occur simultaneously and secondary to the hormonal changes that occur during the ovarian cycle.
- The duration of menstrual cycle is about 28 +/- 3 days.

Phases of the menstrual cycle:

1- *Destructive phase (Menstruation or bleeding)*

- Duration: 3 – 5 days, starting from the first day of menstruation.
- It coincides with the beginning of the *follicular phase* of ovarian cycle.
- Menstrual discharge is about 100 ml (40 ml blood, 35 ml serous fluid, necrosed endometrial tissues and a large number of leucocytes).

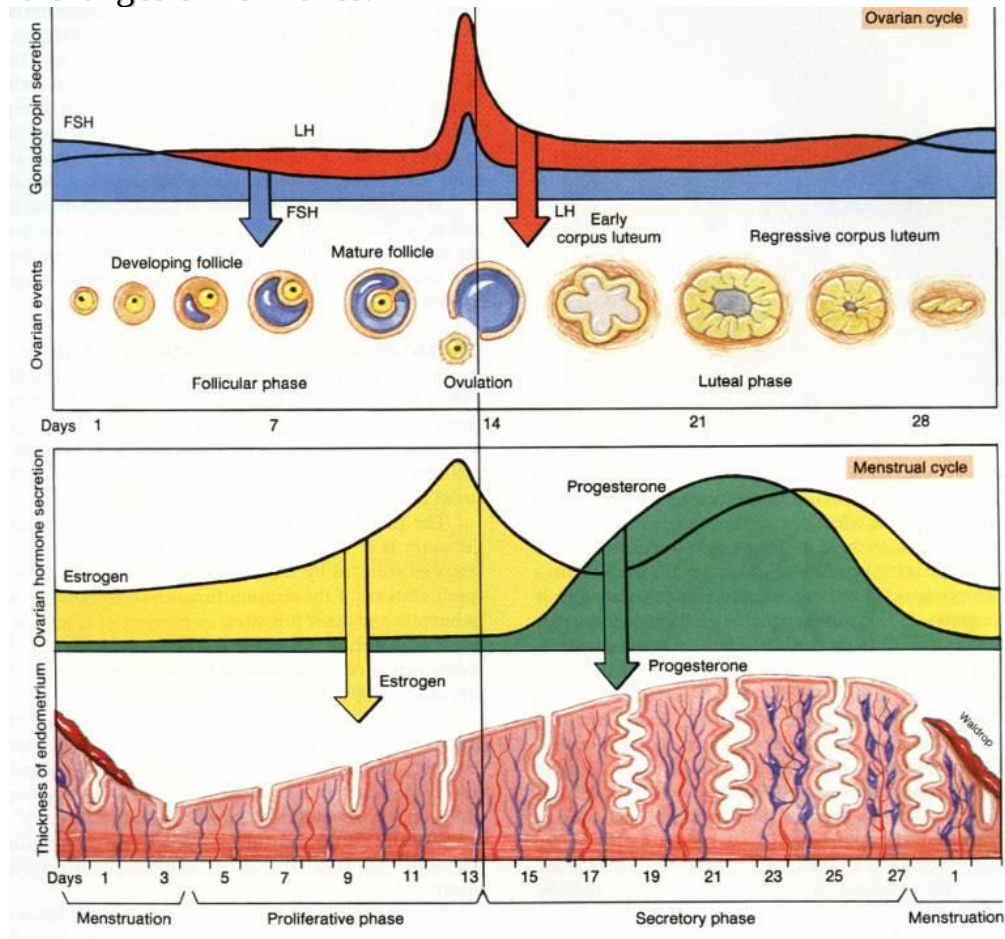
2- *Proliferative phase (Estrogenic phase or Preovulatory phase)*

- Duration: from the 5th day to the 14th day (i.e. from the end of bleeding to ovulation).
- It coincides with the follicular phase of ovarian cycle.
- Endometrial changes are:
- Rapid increase in thickness of Endometrium (3-4 mm thick).
- Uterine glands increase in thickness but, they do not secrete.
- Spiral arteries develop and stroma cells increase in number.
- **Hormonal control:** Proliferative phase is controlled by **estrogens** secreted from growing **ovarian follicle** under the effect of **FSH hormone**.

3- *Secretory phase (progestational, luteal or Postovulatory phase).*

- Duration: from 14th day to the 28th day. (From ovulation to the first day in the next uterine cycle).
- It coincides with the **luteal phase** of ovarian cycle.
- Endometrial changes:
- Endometrial thickness reaches 5 – 6 mm. it becomes highly vascularized and edematous.

- Uterine glands start their secretion.
- **Hormonal control:** Secretory phase is controlled by **progesterone** mainly and **estrogens** secreted from **corpus luteum**.
- The following diagram showing relationship between ovarian cycle & menstrual cycle and changes of hormones.



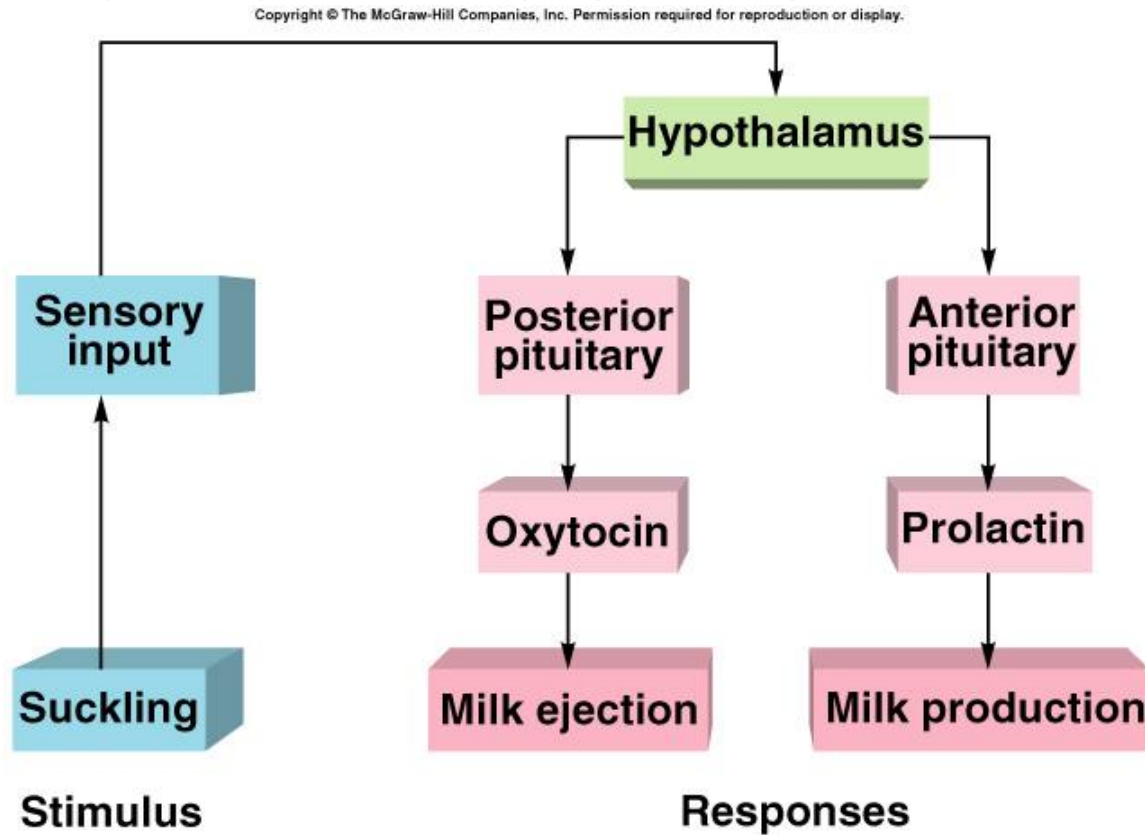
Lactation

Definition: Synthesis and ejection of milk from mammary glands under influence of estrogen, GH, insulin, glucocorticoids, and prolactin

- Breast develop during gestation period
- Stimulated by estrogen (ductal growth), progestin (alveolar proliferation), and prolactin (lactation)
- Progesterone inhibits milk production and secretion, and its abrupt decrease after delivery causes lactation
- 1-3 days postpartum breast secrete **colostrum** (a fluid with same protein & lactose content as milk but less fat, and increase of immunoglobulin).
- **Prolactin** hormone promotes milk synthesis (production)
- Breast feeding does reduce chances of becoming pregnant, but is not always safe
- Menstrual cycles begin again usually within a few weeks to a few months

- Oxytocin hormone has a milk let down effect that allows milk to enter ducts
- Suckling of infant further stimulates milk from breasts, once feeding stops after about a week the gland cease production

Milk-Ejection Reflex: the following diagram showing this reflex



MENOPAUSE

- Loss of all follicles and ovarian estrogen production
- Estrogen may be produced by peripheral aromatization of androstenedione
- Marked increase in LH and FSH
- Atrophy of the urinary tract and vaginal epithelium
- Decrease in bone mass (osteoporosis)

Physiology of reproductive system

The male reproductive system consists of the **1-** primary sex organs (testes). The testes produce sex cells, or gametes and secrete a variety of steroid hormones commonly called sex hormones. **2-** The internal accessory organs of the male reproductive system are the epididymis, vas deferens, seminal vesicles, prostate gland, and bulbourethral glands.

Epididymis functions: receives and stores sperms from seminiferous tubules as these cells are formed. Inside the epididymis, spermatids mature to become sperm cells, absorb excess testicular fluid and to pass nutrients to the many sperm stored temporarily in the lumen. The immature, nearly non-motile sperm that leave the testis are moved slowly along the duct of the epididymis through fluid that contains several kinds of antimicrobial proteins, including defensins. As they move along its tortuous course (a trip that takes about 20 days), the sperm gain the ability to swim.

Vas Deferens: A tube called a vas deferens is connected to each epididymis. These tubes carry sperm cells from an epididymis to the urethra in the pelvic cavity of the male

The Urethra: It conveys both urine and semen (at different times) secrete lubricating mucus into the lumen just before ejaculation and it contain urethral glands that secrete lubricating mucus into the lumen just before ejaculation.

Seminal vesicles: secrete yellowish viscous alkaline fluid containing fructose sugar, citric acid, a coagulating enzyme (vesiculase), and prostaglandins, as well as other substances that enhance sperm motility or fertilizing ability. The sugars are used by sperm cells to make energy, and the prostaglandins stimulate muscular contractions in the female reproductive system. The yellow color of seminal fluid is due to a yellow pigment that fluoresces under UV light, a capability that allows investigators to recognize a sperm trail or residue in instances of sexual attack. Seminal vesicles release their product into the vas deferens just before ejaculation.

Prostate Gland: It makes a milky, alkaline fluid that contains citrate (a nutrient source), several enzymes (including fibrinolysin, hyaluronidase, and acid phosphatase), and prostate-specific antigen (PSA). It secretes this fluid into the urethra just before ejaculation. Also, this fluid helps to protect the sperm when they enter the female reproductive system.

Bulbourethral glands: They produce a thick, clear mucus, some of which drains into the spongy urethra and lubricates the glans penis when a man becomes sexually excited. Additionally, the mucus neutralizes traces of acidic urine and lubricates the urethra just prior to ejaculation.

The reproductive functions of the male can be divided into three major subdivisions:

1. Spermatogenesis, which means simply the formation of sperm.
2. Performance of the male sexual act; regulation of male reproductive functions by the various hormones.
3. Transmission of spermatozoa to the female.
 - Associated with these reproductive functions are the effects of the male sex hormones on the accessory sexual organs, cellular metabolism, growth, and other functions of the body.

Puberty in the male

- Occurs between the ages of 10 and 14
- L.H stimulates the interstitial cells of the testes to increase the production of testosterone
- Testosterone influences the development of the body to sexual maturity

The changes which occur at puberty are:

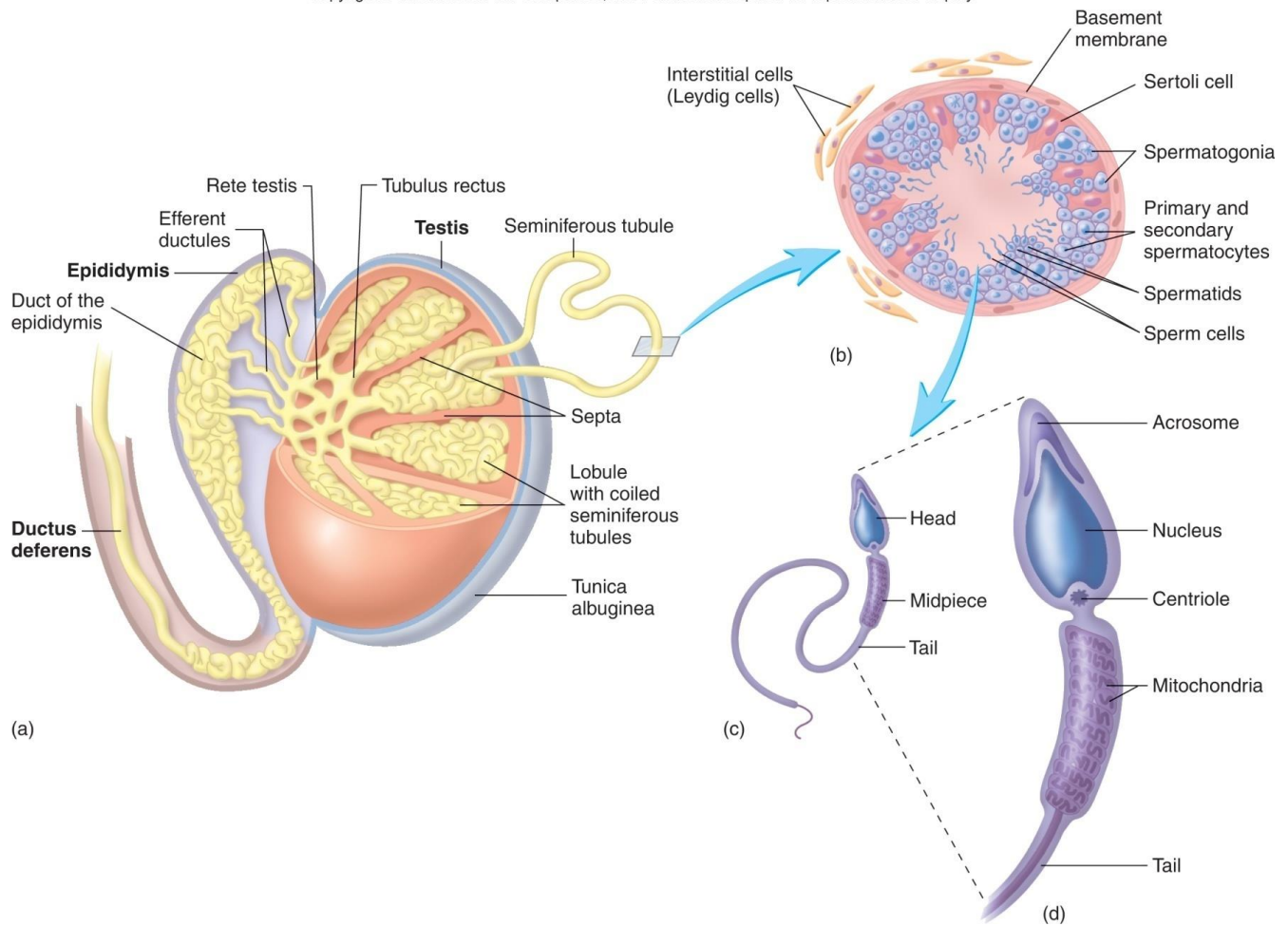
- Growth of muscle, bone and a marked increase in height and weight
 - Enlargement of the larynx and deepening of the voice
 - Growth of hair on the face, chest and abdomen
 - Enlargement of the penis, scrotum and prostate gland
 - Maturation of seminiferous tubules and production of sperm
 - The skin thickens and becomes more oily.
- The male reproductive system consists of the testes, a series of ducts, accessory glands, and supporting structures. The ducts include the epididymides, ductus deferens, also vas deferens), and urethra. Accessory glands include the seminal vesicles, prostate gland, and bulbourethral glands. Supporting structures include the scrotum and penis.

Testis

- The testis are the reproductive glands of the male. Each testis are 200 to 300 lobules and within each lobule are 1-4 seminiferous tubules. Between seminiferous tubules, there are groups of interstitial cells (Leydig cells). Leydig cells secrete testosterone after puberty.

Cell types in the testis:

- 1- Spermatogonia – in seminiferous tubules
- 2- Leydig Cells (Interstitial cells) located in connective tissues between the seminiferous tubules, produce testosterone in response to LH
- 3- Sertoli cells, form the basement membrane of the seminiferous tubules and provide the environment needed for germ cell maturation the following figure showing these cells.



SPERMATOGENESIS means simply the formation of sperm include three phases:

1- Spermatogonial Phase (Mitosis)

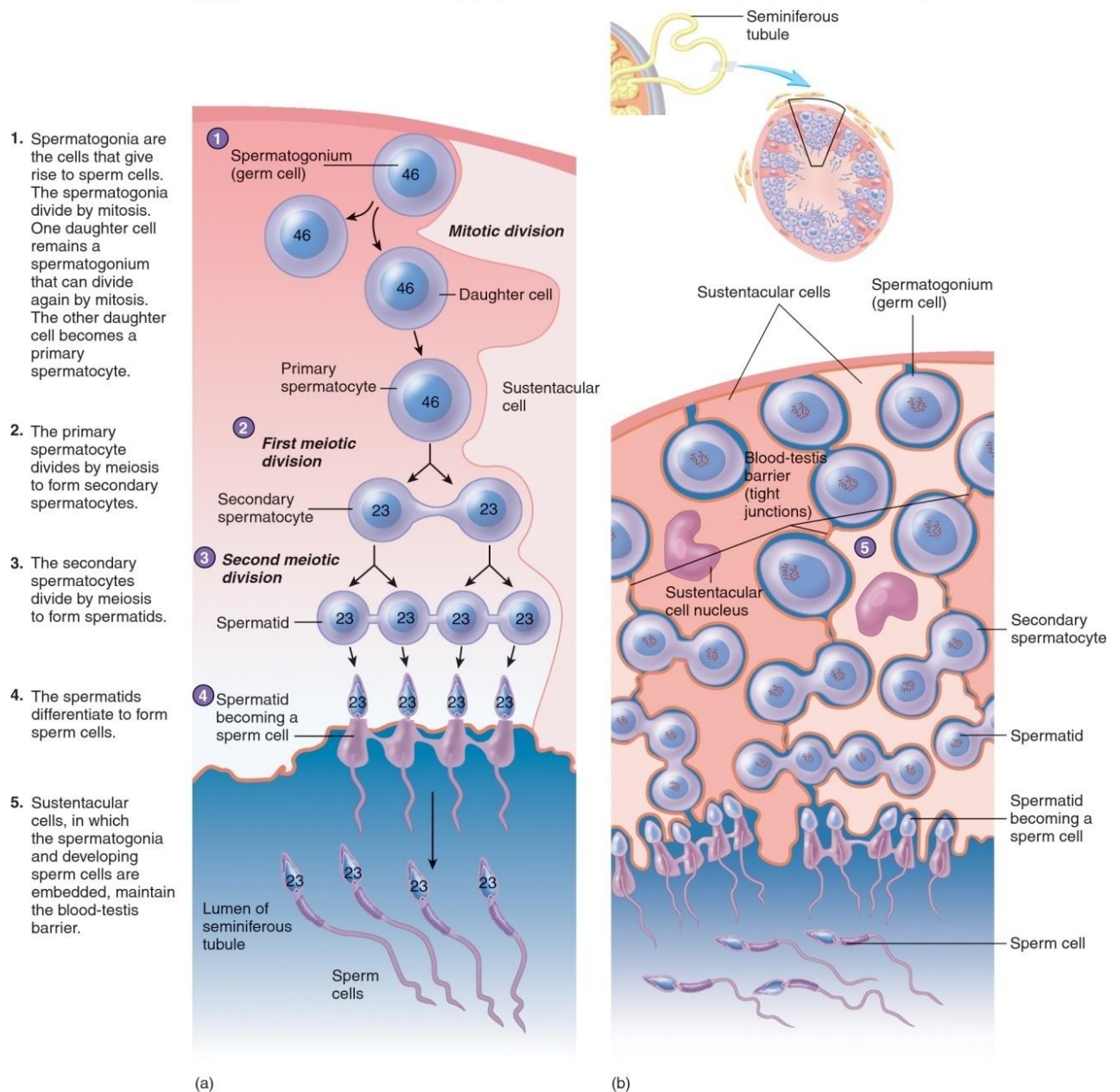
In this phase spermatogonia proliferate by mitotic divisions to provide *stem cells* and cells which, will proceed through spermatogenesis (*1^o spermatocytes*)

2- Spermatocyte Phase (Meiosis) during this phase diploid cells ($2n$) created in spermatogonial phase give rise to haploid cells ($1n$). Meiosis I (reduction division) & Meiosis II (equatorial division) will be occur. *1^o spermatocytes* enter Meiosis I to form *2^o spermatocytes* which then enter Meiosis II and result in spermatids

3- Spermatid Phase (Spermiogenesis) during this phase spermatid differentiation into spermatozoa through *3 steps*:

- a- acrosome formation; golgi granules fuse to form acrosome that contains hydrolytic enzymes which will enable the spermatozoa to move through the investing layers of the oocyte
- b- flagellum formation; centrioles and associate axoneme (arrangement of microtubules in cilia)
- c- changes in size and shape of nucleus; chromatin condenses and shedding of residual body (cytoplasm) the following figure showing spermatogenesis

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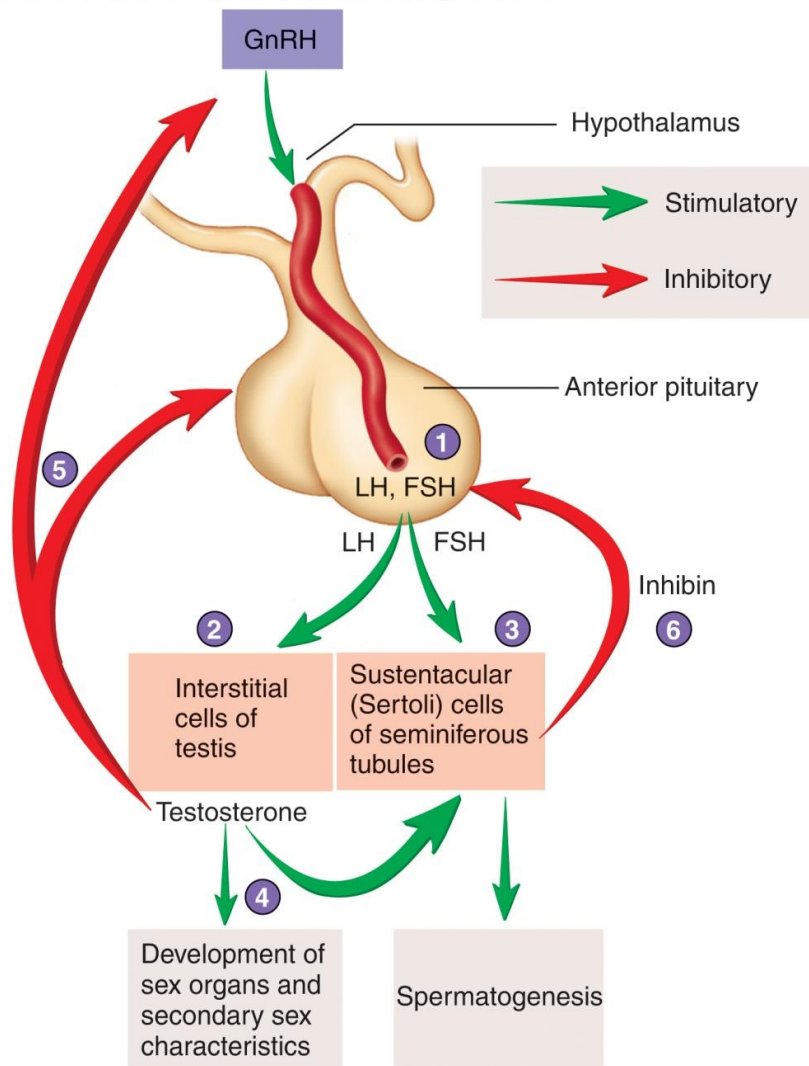
Hormonal Factors That Stimulate Spermatogenesis

- Several hormones play essential roles in spermatogenesis. Some of these are as follows:

1. **Testosterone**, secreted by the *Leydig cells* located in the interstitium of the testis, is essential for growth and division of the testicular germinal cells, which is the first stage in forming sperm.
2. **LH**, secreted by the anterior pituitary gland, stimulates the Leydig cells to secrete testosterone.
3. **FSH**, also secreted by the anterior pituitary gland, FSH enhances the production of androgen-binding protein (ABP) by the Sertoli cells of the testes; androgen binding protein is essential to concentrating testosterone in levels high enough to initiate and maintain spermatogenesis; without this stimulation, the conversion of the spermatids to sperm (the process of spermatogenesis) will not occur.
4. **Inhibin** is a peptide that is an inhibitor of FSH synthesis and secretion. Inhibin is secreted from the sustentacular cell, located in the seminiferous tubule inside the testes
5. **Estrogens**, formed from testosterone by the Sertoli cells when they are stimulated by follicle-stimulating hormone, are probably also essential for spermatogenesis.
6. **Growth hormone** is necessary for controlling background metabolic functions of the testes. Growth hormone specifically promotes early division of the spermatogonia themselves; in its absence, as in pituitary dwarfs, spermatogenesis is severely deficient or absent, thus causing infertility.

The following diagram showing these relationships of hormones and effects.

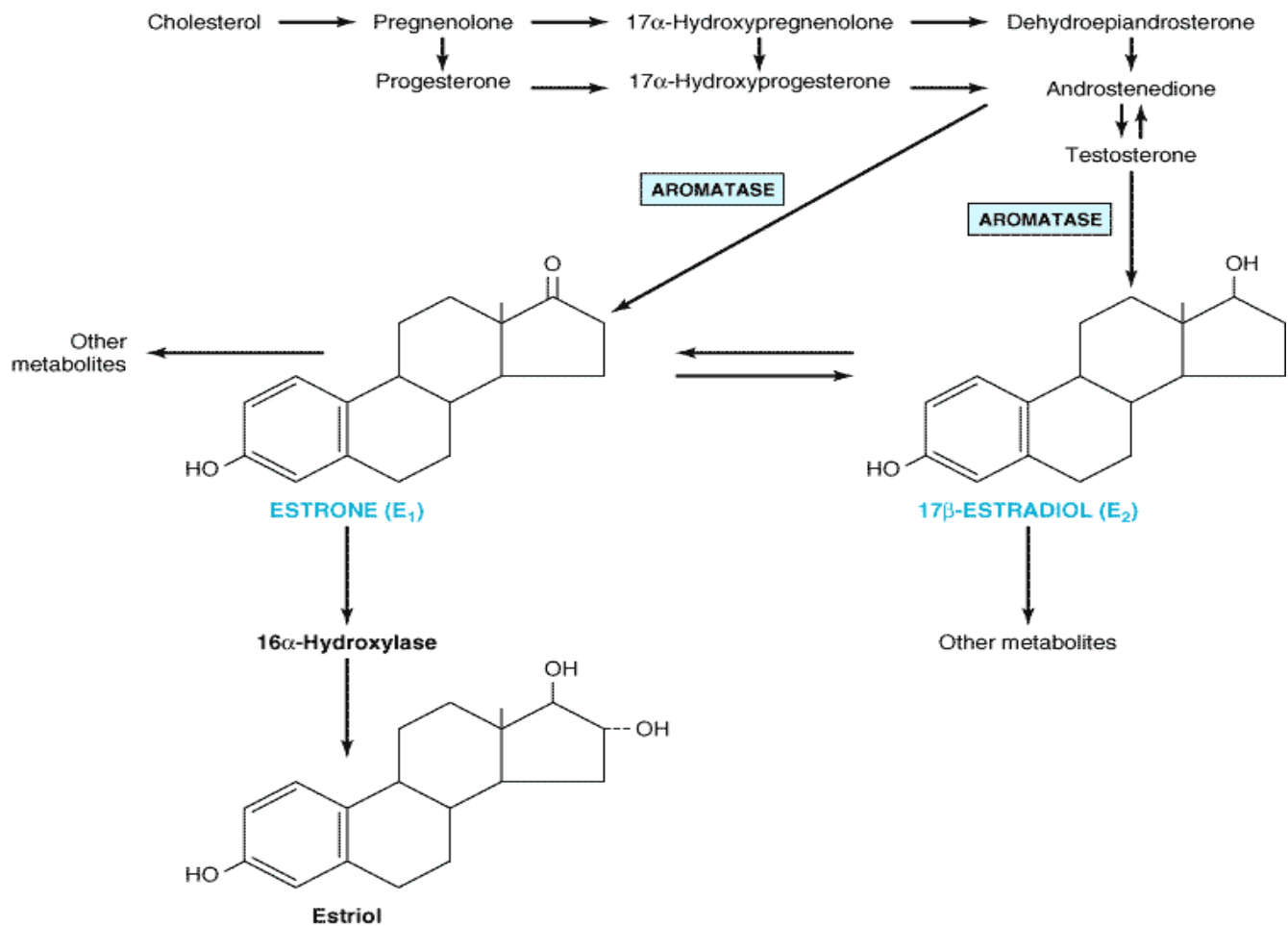
1. GnRH from the hypothalamus stimulates the secretion of LH and FSH from the anterior pituitary.
2. LH stimulates testosterone secretion from the interstitial cells.
3. FSH stimulates sustentacular cells of the seminiferous tubules to increase spermatogenesis and to secrete inhibin.
4. Testosterone has a stimulatory effect on the sustentacular cells of the seminiferous tubules, as well as on the development of sex organs and secondary sex characteristics.
5. Testosterone has a negative-feedback effect on the hypothalamus and pituitary to reduce LH and FSH secretion.
6. Inhibin has a negative-feedback effect on the anterior pituitary to reduce FSH secretion.



Endocrine Function of Testes:

1. Androgens

Testes secrete several male sex hormones, which are collectively called **androgens**, including *testosterone*, *dihydrotestosterone* and *androstenedione*. The testes also produces 17 β Estradiol (E2). Most of the E2 produced by males is derived from peripheral aromatization of testosterone and androstenedione. Testosterone is so much more abundant than the others that one can consider it the significant testicular hormone.



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Functions of testosterone

- 1) Maintenance of spermatogenesis
- 2) Stimulation of development of genitalia.
- 3) Development and maintenance of the male secondary sexual characteristics.
- 4) Maintenance of libido.
- 5) Effect on metabolism.

- Testosterone increases anabolism of protein, especially that in muscle and genitalia, and causes growth of the bones in the thickness, deposition of calcium salts as well enhancement of production of red cells.

Androgen defects

1. Benign Prostatic Hypertrophy (BPH)

- Due to increased DHT or testosterone particularly in older males
- Leads to difficulty in urination due to obstruction of the urethra
- Different from prostatic cancer

2. Hypogonadism

- Decreased testosterone
- May be primary (testicular failure) or secondary (defective gonadotropin secretion)

3. Testicular feminization

- Receptor dysfunction
- Feminized external genitalia